

HKUST MATH 2431 - problem set 9

2.

2. (*) Decay of covariances in an autoregressive process

[4 Points]

Let $\sigma, \alpha \in \mathbb{R}$, $|\alpha| < 1$. We consider random variables X_0 and $(Z_n)_{n \in \mathbb{N}}$, independent and assume that $Z_n \sim \mathcal{N}(0, 1)$. We define a so-called *autoregressive process* (also called AR(1)-process) inductively by

$$X_n = \alpha X_{n-1} + \sigma Z_n, \quad n \in \mathbb{N}.$$

Prove the following claims.

(a) If $X_{n-1} \sim \mathcal{N}(m, v)$, then $X_n \sim \mathcal{N}(\alpha m, \alpha^2 v + \sigma^2)$.

2. a. Norm. dist. are closed under linear combinations, so $X_n \sim \mathcal{N}$.

$$\begin{aligned} E[X_n] &= E[\alpha X_{n-1} + \sigma Z_n] = \alpha E[X_{n-1}] + \sigma E[Z_n] \\ &= \alpha m + \sigma(0) = \alpha m \end{aligned}$$

$$\begin{aligned} \text{Var}(X_n) &= \text{Var}(\alpha X_{n-1} + \sigma Z_n) \\ &= \alpha^2 \text{Var}(X_{n-1}) + \sigma^2 \text{Var}(Z_n) \quad (X_{n-1} \text{ and } Z_n \\ &= \alpha^2 v + \sigma^2(1) = \alpha^2 v + \sigma^2 \quad \text{are independent}) \end{aligned}$$

$$\therefore X_n \sim \mathcal{N}(\alpha m, \alpha^2 v + \sigma^2)$$

(b) The distribution $\mathcal{N}(0, \sigma^2(1 - \alpha^2)^{-1})$ is *stationary*, meaning that

$$X_0 \sim \mu \quad \Rightarrow \quad X_n \sim \mu \text{ for all } n \in \mathbb{N}.$$

$$2.b. \quad X_0 \sim \underbrace{N(0, \sigma^2(1-\alpha^2)^{-1})}_{\mu}$$

Assume $X_n \sim N(\underbrace{0}_{\mu}, \underbrace{\sigma^2(1-\alpha^2)^{-1}}_{\nu})$ for some $n \geq 0$

$$X_{n+1} \sim N(\alpha(0), \alpha^2(\sigma^2(1-\alpha^2)^{-1}) + \sigma^2) \quad \text{by 2.a.}$$

$$= N(0, \frac{\alpha^2\sigma^2 + \sigma^2(1-\alpha^2)}{1-\alpha^2})$$

$$= N(0, \sigma^2(1-\alpha^2)^{-1})$$

By induction, $X_n \sim \mu$ for all $n \in \mathbb{N}$.

(c) Assuming that $X_0 \sim \mu$, show that

$$\text{Cov}[X_n, X_{n-k}] = \alpha^k \frac{\sigma^2}{1-\alpha^2}, \quad 0 \leq k \leq n.$$

2.c. $X_0 \sim \mathcal{N}(0, \sigma^2(1-\alpha^2)^{-1})$,
 we prove ^{the} equivalent statement

$$\text{Cov}[X_n, X_{n+k}] = \alpha^k \frac{\sigma^2}{1-\alpha^2} \quad n, k \geq 0.$$

Fix $n \in \mathbb{N}_0$.

$$\text{Cov}[X_n, X_n] = \text{Var}(X_n) = \frac{\sigma^2}{1-\alpha^2} = \alpha^0 \frac{\sigma^2}{1-\alpha^2}$$

Assume $\text{Cov}[X_n, X_{n+k}] = \alpha^k \frac{\sigma^2}{1-\alpha^2}$ for some $k \in \mathbb{N}_0$

$$\text{Then } \text{Cov}[X_n, X_{n+k+1}] = \text{Cov}[X_n, \alpha X_{n+k} + \sigma Z_n]$$

$$= \alpha \text{Cov}[X_n, X_{n+k}] + \sigma \underbrace{\text{Cov}[X_n, Z_n]}_{=0 \text{ because indep.}}$$

$$= \alpha^{k+1} \frac{\sigma^2}{1-\alpha^2}$$

By induction, $\text{Cov}[X_n, X_{n+k}] = \alpha^k \frac{\sigma^2}{1-\alpha^2}$ for all $k \in \mathbb{N}_0$
 for any $n \in \mathbb{N}_0$

4.

4. (*) Chernoff bounds reloaded

[4 Points]

In this problem you will prove the following theorem:

Let $X_1, \dots, X_n$ independent Bernoulli-distributed random variables with $\mathbf{P}[X_i = 1] = p_i$ and $\mathbf{P}[X_i = 0] = 1 - p_i$. Then, one has for $X = \sum_{i=1}^n X_i$ and $\mu = \mathbf{E}[X] = \sum_{i=1}^n p_i$, and every $\delta > 0$, that $\mathbf{P}[X \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{1+\delta}} \right)^\mu$.

(a) Prove the previous assertion. Use the following steps:

- i. Recall problem 3 on Problem set 7, and apply the result to X .
- ii. Replace X by the sum of X_i . Calculate explicitly the expectation appearing in the formula obtained.
- iii. Replace $1 + p_i(e^t - 1)$ with an upper bound (via $1 + y \leq e^y$) and simplify.
- iv. Find t so that the bound is minimal.

4.a. For X_i , we have

$$\text{for } t > 0 \quad P[X_i \geq \alpha] \leq e^{-t\alpha} E[e^{tX_i}]$$

$$t < 0 \quad P[X_i \leq \alpha] \leq e^{-t\alpha} E[e^{tX_i}]$$

In particular, we have

$$\text{for } t > 0 \quad P[X_i \geq (1+\delta)p_i] \leq e^{-t(1+\delta)p_i} E[e^{tX_i}]$$

Now, consider $X = \sum_i X_i$

$$t > 0, \quad P[X \geq (1+\delta)\mu] \leq e^{-t(1+\delta)\mu} E[e^{tX}]$$

$$E[e^{tX}] = E[e^{t\sum_i X_i}]$$

$$= \prod_i E[e^{tX_i}] \quad \text{because } X_i \text{ are mutually indep.}$$

$$= \prod_i (1 - p_i + p_i e^t)$$

$$= \prod_i (1 + p_i(e^t - 1)) \quad 1 + p_i(e^t - 1) \leq e^{p_i(e^t - 1)}$$

$$\leq \prod_i e^{p_i(e^t - 1)}$$

$$= e^{\mu(e^t - 1)}$$

$$\text{So } P[X \geq (1+\delta)\mu] \leq e^{-t(1+\delta)\mu} e^{\mu(e^t - 1)}$$

$$= e^{\mu(e^t - 1 - t(1+\delta))}$$

$$\min e^{\mu(e^t - 1 - t(1+\delta))}$$

$$\Rightarrow \min e^{t - t(1+\delta)} :$$

$$\text{1st d } e^t - (1+\delta)t : e^t - (1+\delta)t = 0 \Rightarrow t = \ln(1+\delta)$$

$$\text{2nd d } e^t : e^t > 0 \Rightarrow \text{strictly convex}$$

$\ln(1+\delta)$ is min.

$$\text{So } P[X \geq (1+\delta)\mu] \leq e^{-\mu(\ln(1+\delta))}$$

$$\begin{aligned}
 & \leq e^{\mu(1+\delta)} - 1 - (\ln(1+\delta))\mu(1+\delta) \\
 & = e^{\mu(1+\delta)-1} \\
 & = \left(\frac{e^\delta}{(1+\delta)} \right)^\mu (1+\delta)^{-\mu\delta}
 \end{aligned}$$

(note, $\delta > 0$ requires $\ln(1+\delta) > 0 \Rightarrow \delta > 0$,
as expected)

(b) Application: Consider the n -fold toss of a fair coin. Calculate a bound for the probability that the number of "heads" exceeds its expectation by at least 10%, using

- i. the Čebyšev-inequality,
- ii. the version of the Chernoff bound found above,

if $n = 10^3$ or $n = 10^4$. Compare the results.

Hint: Since $p = 1/2$, the distribution is symmetric.

4.b. Using Chebyshev inequality $P(|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}$

$$\mu = (0.5)10^3 = 500$$

$$\sigma = \sqrt{(0.25)10^3} = \sqrt{250}$$

$$\sigma_i^2 = 0.5(1-0.5) = 0.25 \quad \text{mutually indep.}$$

Find $P(X-\mu \geq 50)$.

Symmetry arg implies the left and right tail distributions around the mean are the same.

$$\text{So } P(X-\mu \geq k\sigma) \leq \frac{1}{2k^2}$$

$$P(X-\mu \geq 50) \quad k = \frac{50}{\sqrt{250}} = \sqrt{10} \\ \leq \frac{1}{20}$$

$$n=10^3$$

$$n=10^4$$

$$\mu = (0.5)10^4 = 5000$$

$$\sigma = \sqrt{(0.25)10^4} = \sqrt{2500} = 50$$

Find $P(X-\mu \geq 500)$.

$$k = \frac{500}{50} = 10$$

$$P(X-\mu \geq 500) \\ \leq \frac{1}{200}$$

Using the Chernoff bounds results above

$$P[X \geq (1+\delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{(1+\delta)}} \right)^\mu$$

$$\mu = 500$$

$$n=10^3$$

$$P[X \geq 1.1\mu] \quad \delta = 0.1$$

$$\leq \left(\frac{e^{0.1}}{1.1^{1.1}} \right)^{500}$$

$$\approx 0.0888683$$

$$n=10^4$$

$$\mu = 5000$$

$$P[X \geq 1.1\mu] \quad \delta = 0.1$$

$$\leq \left(\frac{e^{0.1}}{1.1^{1.1}} \right)^{5000}$$

$$\approx 3.07236 \times 10^{-11}$$

Comparisons:

$n=10^3$ vs $n=10^4$, $n \uparrow \Rightarrow$ prob of exu $\rightarrow \downarrow$
bound (i) vs bound (ii), (ii) is much more tighter than (i).

$$(ii) \text{ bound } \propto e^{-n}$$

$$(i) \text{ bound } \propto \frac{1}{n}$$

So (ii) is much better for bounding than (i), and
bounds prob decreases much faster.

this is because bound (i) works for any dist., including
heavy tailed ones, so the bound is weak, whereas
for bound (ii), it works for n independent Bernoulli
trials, so the bound is much stronger.